Module 06 Lab - Regression and Classification Models**Objective:** To understand the difference between regression and classification and to build your first linear models for both tasks.**In this lab, you will write the code to train and evaluate the models.**

Part 1: Regression vs. ClassificationThis is the most fundamental distinction between types of supervised learning problems.* **Regression:** The goal is to predict a **continuous numerical value**. * *Examples:* Predicting the price of a house, the temperature tomorrow, or the stock price.* **Classification:** The goal is to predict a **discrete category or class label**. * *Examples:* Predicting if an email is spam or not spam, if a flower is a setosa, versicolor, or virginica, or if a customer will churn or not.**In this lab, we will tackle one of each.**

Part 2: Linear Regression**Concept:** Linear Regression is used to predict a continuous value. It works by finding the best-fitting straight line through the data points. The model learns a "slope" (coefficient) for each feature and an "intercept".* **Problem:** We will predict the `Fare` of a Titanic passenger based on their `Age` and `Pclass`.

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import train_test_split

# Load and prepare the data
df = pd.read_csv(
    'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.
)

# For simplicity, we'll drop rows with missing age
df.dropna(subset=['Age'], inplace=True)

# Define features and target
features = ['Age', 'Pclass']
target_reg = 'Fare'

X_reg = df[features]
y_reg = df[target_reg]

# Split the data
X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(
    X_reg, y_reg, test_size=0.2, random_state=42
)
```

Task 1: Train and Evaluate a Linear Regression Model**Your Task:**1. Create an instance of the `LinearRegression` model.2. Train the model using the training data (`X_train_reg`, `y_train_reg`).3. Make predictions on the test data.4. Evaluate the model using `mean_squared_error`. This metric tells us the average of the squared differences between the predicted and actual values.

```
# 1. Create the model instance
lr_model = LinearRegression()

# 2. Train the model
lr_model.fit(X_train_reg, y_train_reg)

# 3. Make predictions
y_pred_reg = lr_model.predict(X_test_reg)

# 4. Evaluate the model
mse = mean_squared_error(y_test_reg, y_pred_reg)
print(f"Mean Squared Error for Fare Prediction: {mse:.2f}")
```

```
print(
    f"The square root of this is {np.sqrt(mse):.2f}, meaning our model is off"
    f" by about ${np.sqrt(mse):.2f} on average."
)
```

```
Mean Squared Error for Fare Prediction: 3364.92
The square root of this is 58.01, meaning our model is off by about $58.01 on av
```

Part 3: Logistic Regression**Concept:** Despite its name, Logistic Regression is a **classification** algorithm. It works by calculating the probability that a given input belongs to a certain class. It's one of the most widely used and interpretable classification models.* **Problem:** We will predict whether a passenger `Survived` based on their `Age`, `Pclass`, and `Sex`.

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Prepare data for classification: encode 'Sex' column
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Define features and target
features_cls = ['Age', 'Pclass', 'Sex']
target_cls = 'Survived'

X_cls = df[features_cls]
y_cls = df[target_cls]

# Split the data
X_train_cls, X_test_cls, y_train_cls, y_test_cls = train_test_split(
    X_cls, y_cls, test_size=0.2, random_state=42
)
```

Task 2: Train and Evaluate a Logistic Regression Model**Your Task:**1. Create an instance of the `LogisticRegression` model.2. Train the model using the classification training data.3. Make predictions on the test data.4. Evaluate the model using `accuracy_score`.

```
# 1. Create the model instance
log_model = LogisticRegression()

# 2. Train the model
log_model.fit(X_train_cls, y_train_cls)

# 3. Make predictions
y_pred_cls = log_model.predict(X_test_cls)

# 4. Evaluate the model
accuracy = accuracy_score(y_test_cls, y_pred_cls)
print(f"Accuracy for Survival Prediction: {accuracy:.2%}")
```

```
Accuracy for Survival Prediction: 74.83%
```

📝 Knowledge Check**Instructions:** Answer the following questions in this markdown cell.1. **In your own words, what is the key difference between a regression problem and a classification problem?**2. **The `LinearRegression` model has an attribute called `.coef_`. After you train the model, print `lr_model.coef_`. What do these numbers represent?**3. Why did we use `mean_squared_error` to evaluate the regression model but `accuracy_score` for the classification model? Why wouldn't accuracy be a good metric for the fare prediction task?**[Regression vs. Classification:

Regression predicts continuous values (e.g., fare price).

Classification predicts discrete categories (e.g., survived vs. died).

Meaning of lr_model.coef_:

These are the weights/slopes for Age and Pclass. Each coefficient shows how much the predicted fare changes when that feature increases by 1 unit, keeping the other feature constant.

Evaluation Metrics Selection:

MSE measures how far numerical predictions deviate from actual dollar values.

Accuracy only checks for exact matches. Since continuous fares are almost never identical to the penny, accuracy would be practically 0%.]**